

PROJECT SUTRA — TWIN GRAND FINALS PITCH SCRIPT

STRICT CO-PRESENTER FORMAT: NIKHIL (TECH LEAD) & HARIKA (OPERATIONS & AUDIT LEAD)

PRESENTERS: NIKHIL • HARIKA
TOTAL TIME: ~2 MIN 30 SEC

SLIDE 01 Introduction: Project SUTRA

00:00 - 00:35 (35s)

SPEAKER: NIKHIL (TECH LEAD) [STAGE PRESENCE: CENTER-LEFT • DIRECT JURY EYE CONTACT]

HOOK: "When a disaster strikes, the first 72 hours decide who lives. But the moment drones enter thick forests or mountain ravines, they go blind."

"Respected jury, we are **Team OFFGRID**. I am Nikhil, Tech Lead, and with me is Harika. We built **Project SUTRA** — an autonomous drone swarm that finds survivors in GPS-denied, zero-network disaster zones."

"Instead of single manual drones, SUTRA deploys an intelligent swarm backed by 3 breakthrough pillars:"

50Hz GNC Decentralized Flight: Drones coordinate locally at 50Hz without ground towers.

-5dB JSCC Neural Video: Feeds stay connected through mountain ridges without blackouts.

3.59cm GEO 3D Terrain AI: Pinpoints survivors on steep slopes with centimeter accuracy.

HANDOFF TO HARIKA → "Harika will show why traditional search and rescue collapses in the field."

SLIDE 02 The Problem: 4 Critical Failure Voids

00:35 - 01:20 (45s)

SPEAKER: HARIKA (OPERATIONS & AUDIT LEAD) [STAGE PRESENCE: STEP FORWARD • CONFIDENT & FACTUAL]

HOOK: "Thank you, Nikhil. When we audited actual NDRF disaster reports, we found 4 fatal engineering voids that cause current missions to fail:"

SUBSYS A (GNC) Canopy GPS Drift: Trees block satellite signals. Drones drift and crash into branches. (70% loss in Wayanad).

SUBSYS B (COMMS) Mountain RF Blackout: Ridges cut radio line-of-sight. Standard video drops off a digital cliff below 5dB SNR.

SUBSYS C (VISION) Flat-Earth AI Error: 2D cameras assume flat ground. On steep slopes, coordinates miss by **15 to 30 meters**.

SUBSYS D (C2 GCS) Pilot Bottleneck: Requires 1 pilot per drone and **15 to 25 crew**, costing ₹12.5 Lakhs per deployment.

HANDOFF TO NIKHIL → "Nikhil, walk the jury through how SUTRA's architecture solves all four voids."

SLIDE 03 The Solution: 4 Architectural Moats & 3-4x Performance Delta

01:20 - 02:25 (65s)

NIKHIL: CORE TECH MOATS (A & B) [25s • DEEP TECHNICAL AUTHORITY]

"SUTRA answers each void with a hardened, defensible technical moat:"

MOAT A (GNC) SUTRA-FSD & 3D ORCA: Onboard 50Hz obstacle avoidance. Autonomous flight with zero towers or GPS.

MOAT B (COMMS) Deep JSCC Neural Video: Analog neural transmission. Clear video down to -5dB SNR with zero digital cliff.

HARIKA: VISION, C2 GCS & MISSION BENCHMARKS [40s • CLOSING PUNCH & ROI]

MOAT C (VISION) 3D DEM AI Geolocation: Edge YOLOv8 + 3D terrain raycasting on Jetson Orin. **3.59cm accuracy** in <15ms.

MOAT D (C2 GCS) Pegasus 3D GCS: Offline tactical twin. **1 to 2 operators fly the whole swarm** for just **₹95,000 / sortie**.

"The result? SUTRA scans a square mile in **10 to 18 minutes** — **3 to 4 times faster than traditional teams**, saving lives when every second counts."

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min (3-4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000